

E-Commerce Management System

Module: Cart Management Module — End User & Technical Documentation

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GitHub Repo: ansarizeenat/ecommerce-category-management	Live Server (Render): https://ecommerce-category-management.onrender.com
Tech Stack: Java 25, Spring Boot 4.1, JPA, MySQL 8.0, Thymeleaf	Date: September 2026

1. Introduction

The **Cart Management Module** allows customers to add, update, and remove products from the shopping cart. It ensures that customers can review their selected items before proceeding to checkout, updating prices and quantities in real time. Inventory is validated for every change so the shop never oversells. Admins have a dashboard to view any customer's cart and to analyse cart-abandonment patterns.

2. Database Design & Specification

Table Name: **carts** — schema matches the Itvedant assignment specification.

Table 1: carts

Column	Data Type	Key / Constraint	Description
cart_id	INT	PK, Auto-Increment	Unique identifier for each cart line item
customer_id	INT	FK (users.user_id)	Customer who owns this cart row (shopper)
product_id	INT	FK (products.id)	Product added to the cart
quantity	INT	NOT NULL	Number of units of the product in this row
total_price	DECIMAL(10,2)	NOT NULL	Line total (unit price × quantity)
created_at	DATETIME	AUTO-POPULATED	Timestamp when the item was added to cart
updated_at	DATETIME	AUTO-UPDATED	Timestamp when the cart row was last modified

3. Core Functionalities

1. Add to Cart: Customers can add products to the shopping cart directly from the product page. The system validates that the product is active and in stock before saving. If the same product is already in the customer's cart, the quantities are **merged** instead of creating a duplicate row.

2. Update Cart: Customers can modify the quantity of any line item in the cart and the total price is updated accordingly in real time. The system re-checks inventory for the new quantity to prevent oversell.

3. Remove from Cart: Customers remove individual items; the total is recalculated. Admins can also clear an entire customer's cart in a single click (useful for abandoned cart or support).

4. Cart Dashboard: Customers see their cart contents (product, SKU, unit price, qty, line total, stock). Admins view all customers' carts and use summary stats for cart-abandonment analysis: Total Cart Value, Customers With Carts, Total Items Quantity, and Avg Items / Customer.

4. End User Steps

1. Open <https://ecommerce-category-management.onrender.com> and select the **Cart Management** tab.
2. Ensure a customer and an in-stock product exist (create them via the Customer / Product tabs if needed).
3. Choose a customer, pick a product, enter a quantity and click **Add to Cart**.
4. Use the **Qty** pencil button to change quantities, or the trash icon to remove a line.
5. Use the customer filter to select a specific customer and then click **Clear Cart** to wipe their entire cart.
6. Read the four dashboard summary cards to analyse total cart value and cart-abandonment patterns.

5. REST API Endpoint Specifications

Method	Endpoint	Description	Sample Payload / Response
POST	<code>/api/carts</code>	Add product to cart (merges with existing)	<code>{"customerId":1, "productId":5, "quantity":2}</code>
GET	<code>/api/carts</code>	All carts (use <code>?customerId=X</code> to filter)	JSON list of cart rows
GET	<code>/api/carts/stats</code>	Cart-abandonment summary stats	totalCartValue, uniqueCustomers, totalQty, avgItems
GET	<code>/api/carts/customer/{id}</code>	Get one customer's cart	Filtered cart list
GET	<code>/api/carts/customer/{id}/summary</code>	Customer cart grand total + counts	grandTotal, uniqueItems, totalQuantity
PUT	<code>/api/carts/{id}/quantity</code>	Update qty (re-checks stock)	<code>{"quantity":3}</code>
DELETE	<code>/api/carts/{id}</code>	Remove one cart line item	Success message
DELETE	<code>/api/carts/customer/{id}</code>	Clear ENTIRE cart for a customer	Success message

6. Automated Test Suite

Tests in `CartServiceTest.java` cover: add success, qty merge for duplicate product + customer, insufficient stock rejection, merge-exceeds-stock rejection, quantity update + price recalc, update-exceeds-stock, remove item, admin stats + customer summary, clear-customer-cart, and inactive-product blocking.

7. Assignment Submission Summary

Item	Details & Links
1. GitHub Repository	https://github.com/ansarizeenat/ecommerce-category-management
2. Free Cloud Hosted URL	https://ecommerce-category-management.onrender.com
3. End User Documentation	This PDF — Cart Management Module End User Documentation.
4. Modules Completed	Category, Product, Order, Customer, Payment, and Cart Management.